

VMMC Symbolic Transition Matrix: Representative Entries

Source state $\mathbf{s} = \{(1, 1)^1, (1, 2)^2, (1, 3)^3\}$

SZ-DBC checker — BFS output, `vmmc_2d.w1`, $n_{\text{grid}} = 3$

System and notation

The source state for all entries is orbit representative 1:

$$\mathbf{s} = \underbrace{\{(1, 1), 1\}}_{\text{pos}}, \{(1, 2), 2\}, \{(1, 3), 3\}$$

three distinct-type particles in the first row of a 3×3 torus. Each transition entry $T(\mathbf{s} \rightarrow \mathbf{t})$ is the sum of leaf weights from the BFS tree rooted at $\mathbf{s}$.

Coupling constants. $\mathcal{J}_{ab,r}$ denotes `couplingJ[a,b,r]`: the bare pair energy between particle types a and b at squared lattice distance $r \in \{1, 2\}$. There are 12 symbolic atoms $\{\mathcal{J}_{11,1}, \mathcal{J}_{11,2}, \mathcal{J}_{12,1}, \mathcal{J}_{12,2}, \mathcal{J}_{13,1}, \mathcal{J}_{13,2}, \mathcal{J}_{22,1}, \mathcal{J}_{22,2}, \mathcal{J}_{23,1}, \mathcal{J}_{23,2}, \mathcal{J}_{33,1}, \mathcal{J}_{33,2}\}$.

VMMC link probabilities. The Whitelam–Geissler cluster builder tests each candidate neighbour q against two bond probabilities.

- **Forward bond** (energy rises when p moves away from q):

$$W_{ab} = \begin{cases} 1 - e^{\beta(\mathcal{J}_{ab,1} - \mathcal{J}_{ab,2})} & \text{if } \mathcal{J}_{ab,1} < \mathcal{J}_{ab,2}, \\ 0 & \text{otherwise.} \end{cases}$$

Write $\bar{W}_{ab} = 1 - W_{ab}$.

- **Reverse-only bond** (energy rises when p would move *toward* q in the reverse direction, with no current interaction):

$$R_{ab} = \begin{cases} 1 - e^{\beta \mathcal{J}_{ab,1}} & \text{if } \mathcal{J}_{ab,1} < 0, \\ 0 & \text{otherwise.} \end{cases}$$

Write $\bar{R}_{ab} = 1 - R_{ab}$.

Base factor. Selecting one particle from three (2 bits via `RandomChoice`) and one direction from eight (3 bits via `RandomChoice`) gives a base factor of $(1/2)^2 \times (1/2)^3 = 1/32$ per leaf path. A denominator of $1/16$ indicates two symmetry-related paths (same bond structure, different particle/direction) that contribute identical Piecewise expressions and are summed by the BFS.

Piecewise conditions in rep 1. The BFS over $\mathbf{s}$ finds exactly **six distinct branch conditions**:

$$\underbrace{(\mathcal{J}_{12,1} < \mathcal{J}_{12,2}), (\mathcal{J}_{13,1} < \mathcal{J}_{13,2}), (\mathcal{J}_{23,1} < \mathcal{J}_{23,2})}_{\text{forward-bond conditions } A_{ab}} \quad \text{and} \quad \underbrace{(\mathcal{J}_{12,1} < 0), (\mathcal{J}_{13,1} < 0), (\mathcal{J}_{23,1} < 0)}_{\text{reverse-bond conditions } B_{ab}}$$

This gives $k = 6$, i.e. $2^6 = 64$ branch regions to test under EBE.

1 Entries 1–3: single-bond moves (depth 6, one leaf each)

These are the simplest non-trivial transitions: one particle moves in a direction that takes it away from exactly one current neighbour. The VMMC builder tests only that one bond; the other particles are too far from the moved particle’s new position to be included in the cluster.

Entry 1: $\mathbf{s} \rightarrow \{(1,1)^1, (1,2)^2, (2,1)^3\}$ — **particle 3 moves**

$$T(\mathbf{s} \rightarrow \mathbf{t}_1) = \frac{\bar{W}_{23}}{32} = \frac{1}{32} \left(1 - W_{23} \right) = \frac{1}{32} \begin{cases} e^{\beta(\mathcal{J}_{23,1} - \mathcal{J}_{23,2})} & \text{if } \mathcal{J}_{23,1} < \mathcal{J}_{23,2}, \\ 1 & \text{otherwise.} \end{cases}$$

Bit sequence depth: 6 (5 selection bits + 1 bond-acceptance bit).

Physical meaning: particle 3 moves diagonally away from particle 2 (type-2,3 pair). The forward bond probability W_{23} is non-zero only when $\mathcal{J}_{23,1} < \mathcal{J}_{23,2}$, i.e. when the energy increases as the pair separates. The leaf weight captures the probability that the bond was *not* formed ($1 - W_{23}$), so particle 3 moves alone.

Entry 2: $\mathbf{s} \rightarrow \{(1,1)^1, (1,2)^2, (2,2)^3\}$ — **particle 3 moves (different direction)**

$$T(\mathbf{s} \rightarrow \mathbf{t}_2) = \frac{\bar{W}_{13}}{32} = \frac{1}{32} \begin{cases} e^{\beta(\mathcal{J}_{13,1} - \mathcal{J}_{13,2})} & \text{if } \mathcal{J}_{13,1} < \mathcal{J}_{13,2}, \\ 1 & \text{otherwise.} \end{cases}$$

Depth: 6. Only the type-1,3 bond is tested (particle 1 at (1,1) is now the relevant close neighbour after the move).

Entry 3: $\mathbf{s} \rightarrow \{(1,1)^1, (1,2)^2, (3,1)^3\}$ — **same expression as Entry 1**

$$T(\mathbf{s} \rightarrow \mathbf{t}_3) = \frac{\bar{W}_{23}}{32}$$

Depth: 6. Despite a different target state (particle 3 moved further, to row 3), the single bond tested is still the 2,3 pair at the same distance class change, giving the identical Piecewise expression. This illustrates that distinct transitions can carry identical symbolic weights.

2 Entries 4–5: two-bond moves (depth 7, one leaf each)

When the moved particle separates from two neighbours simultaneously, the BFS generates one additional branch, giving depth 7.

Entry 4: $\mathbf{s} \rightarrow \{(1,1)^1, (1,2)^2, (2,3)^3\}$ — **particle 3, two bonds checked**

$$T(\mathbf{s} \rightarrow \mathbf{t}_4) = \frac{\bar{W}_{13} \cdot \bar{W}_{23}}{32}$$

Depth: 7. Particle 3 moves into a position where both particle 1 (type 1,3 pair) and particle 2 (type 2,3 pair) are candidates. Neither bond is formed: $T(\mathbf{s} \rightarrow \mathbf{t}_4)$ is non-zero in all four branch regions of $(A_{13}, A_{23}) \in \{0,1\}^2$, but takes different values in each.

Entry 5: $\mathbf{s} \rightarrow \{(1,1)^1, (1,3)^3, (2,2)^2\}$ — **particle 2 moves**

$$T(\mathbf{s} \rightarrow \mathbf{t}_5) = \frac{\bar{W}_{12} \cdot \bar{W}_{23}}{32}$$

Depth: 7. Now particle 2 (type 2) moves away from both particle 1 (type 1,2 pair) and particle 3 (type 2,3 pair). The bond conditions now involve A_{12} and A_{23} rather than A_{13} and A_{23} . This entry demonstrates that different particle choices activate different pairs of branch conditions from the $k = 6$ total.

3 Entry 6: the self-loop — 54 leaves, depth 6–10

The self-loop $T(\mathbf{s} \rightarrow \mathbf{s})$ collects all paths that return to $\mathbf{s}$: rejected moves, frustrated cluster builds, and overlap-blocked moves. It has **54 contributing leaves** at depths 6–10. A representative sample of six terms is shown, illustrating the three structural classes:

Class A — bond formed then frustrated (depth 6). A forward bond is accepted (W_{ab} factor) but the frustration check immediately rejects the cluster because the reverse bond probability is zero:

$$(i) \quad \frac{W_{12}}{32} = \frac{1}{32} \begin{cases} 1 - e^{\beta(\mathcal{J}_{12,1} - \mathcal{J}_{12,2})} & \mathcal{J}_{12,1} < \mathcal{J}_{12,2}, \\ 0 & \text{o.w.} \end{cases} \quad (1)$$

$$(ii) \quad \frac{W_{13}}{32} = \frac{1}{32} \begin{cases} 1 - e^{\beta(\mathcal{J}_{13,1} - \mathcal{J}_{13,2})} & \mathcal{J}_{13,1} < \mathcal{J}_{13,2}, \\ 0 & \text{o.w.} \end{cases} \quad (2)$$

$$(iii) \quad \frac{W_{23}}{32} \quad (\text{analogously for the 2,3 pair}) \quad (3)$$

Class B — reverse-bond check, no cluster formed (depth 6). The moved particle has no current interaction with a candidate neighbour, but the reverse bond check (condition B_{ab}) contributes:

$$(iv) \quad \frac{\bar{R}_{13}}{32} = \frac{1}{32} \begin{cases} e^{\beta \mathcal{J}_{13,1}} & \mathcal{J}_{13,1} < 0, \\ 1 & \text{o.w.} \end{cases} \quad (4)$$

$$(v) \quad \frac{\bar{R}_{12}}{32}, \quad \frac{\bar{R}_{23}}{32} \quad (\text{analogously for the other pairs}) \quad (5)$$

Class C — multi-bond frustrated path (depth 9–10). Two bonds are tested and accepted, but the frustration check on the second neighbour rejects the cluster. A representative depth-9 term:

$$(vi) \quad \frac{1}{16} \underbrace{\theta_{12}^+}_{=1 \text{ iff } A_{12}} \cdot \underbrace{(1 - \theta_{13}^+)}_{=1 \text{ iff } \neg A_{13}} \cdot W_{12} \cdot W_{13} \cdot \bar{W}_{23} \quad (6)$$

where $\theta_{ab}^+ \equiv \mathbf{1}[\mathcal{J}_{ab,1} < \mathcal{J}_{ab,2}]$ is the indicator of the forward-bond condition. The factor $1/16 = 2 \times (1/32)$ reflects two symmetry-related particle/direction choices giving identical weights.

The full $T(\mathbf{s} \rightarrow \mathbf{s})$ expression contains **22 distinct terms**, spanning all six conditions $\{A_{12}, A_{13}, A_{23}, B_{12}, B_{13}, B_{23}\}$ and products thereof. It is non-zero for every coupling-constant configuration (since the cluster builder always has a non-zero probability of rejection), and its complexity highlights why the self-loop is the hardest single entry to evaluate symbolically.

4 Entries 7–10: 3-particle cluster move — 9 leaves, depth 9–10

When all three particles move together, the cluster builder must accept *two* bond pairs successfully. With three particles there are three distinct ways to “chain” them: (1,2) and (1,3); (1,2) and (2,3); (1,3) and (2,3).

The target state $\mathbf{t}_6 = \{(2,1)^1, (2,2)^2, (2,3)^3\}$ (all particles shifted to row 2) has exactly 9 contributing leaves. We display six of them:

Entries 7–9: three depth-9 “two-bond” leaves

Each of the three chaining routes contributes one depth-9 leaf. The seed particle opens a bond with one neighbour (accepted), that neighbour then opens a bond with the third particle (accepted), and no further neighbours are in range:

$$\ell_1 : \quad \frac{1}{32} \theta_{12}^+ \theta_{13}^+ W_{12} W_{13} \quad (7)$$

$$\ell_2 : \quad \frac{1}{32} \theta_{12}^+ \theta_{23}^+ W_{12} W_{23} \quad (8)$$

$$\ell_3 : \quad \frac{1}{32} \theta_{13}^+ \theta_{23}^+ W_{13} W_{23} \quad (9)$$

The indicator pairs $(\theta_{12}^+, \theta_{13}^+)$ etc. are products of indicator functions that are themselves Piecewise with value 1 inside the respective branch region and 0 outside. Each leaf is **active in only one quadrant** of the (A_{ab}) space: ℓ_1 is non-zero only when $A_{12} \wedge A_{13}$, etc.

Entry 10: a depth-10 leaf with three-bond check

When all three bond conditions are active simultaneously, the BFS explores a third candidate after the second bond is accepted, contributing depth-10 leaves that include a “check and reject” factor $(1 - W_{ab})$:

$$\ell_4 : \quad \frac{1}{16} \theta_{13}^+ \theta_{23}^+ \underbrace{(1 - W_{12})}_{\text{12 bond NOT formed}} \cdot W_{13} \cdot W_{23} \quad (10)$$

The factor $1/16$ (doubling of $1/32$) reflects two particle/direction paths that both trigger this three-candidate sequence.

The full $T(\mathbf{s} \rightarrow \mathbf{t}_6)$ is the sum $\ell_1 + \ell'_1 + \ell_2 + \ell'_2 + \ell_3 + \ell'_3 + \ell_4 + \ell_5 + \ell_6$ where primed terms are depth-10 analogues of ℓ_1, ℓ_2, ℓ_3 that include one extra “third neighbour checked and rejected” factor.

The target state $\mathbf{t}_7 = \{(3, 1)^1, (3, 2)^2, (3, 3)^3\}$ (all shifted to row 3) has **exactly the same Piecewise expression** as $T(\mathbf{s} \rightarrow \mathbf{t}_6)$. This is a direct consequence of translational invariance: the row-3 target is the row-2 target translated by one row, and the BFS correctly produces identical symbolic transition weights in both cases.

5 Discussion: structure and implications for EBE

5.1 What the expressions reveal

Only 6 distinct conditions for this orbit rep. Despite a BFS depth of up to 10 and 102 total leaves, the entire transition row from $\mathbf{s}$ involves only $k = 6$ distinct Piecewise branch conditions: three forward-bond conditions A_{ab} and three reverse-bond conditions B_{ab} . The coupling-constant space is partitioned into at most $2^6 = 64$ branch regions.

Each leaf activates a specific subset of conditions. Shallow leaves (depth 6) activate exactly one condition. Medium leaves (depth 7) activate a product of two. Deep leaves (depth 9–10) activate three or more. No single leaf activates all six simultaneously: the depth-10 leaves involve at most four conditions at once.

The self-loop mixes forward and reverse conditions. Leaves contributing to $T(\mathbf{s} \rightarrow \mathbf{s})$ include both A_{ab} and B_{ab} conditions because frustrated paths can arise either from an active forward bond with no reverse partner (class A) or from a reverse-only bond with no forward partner (class B). This is the only transition row that exercises the B_{ab} conditions.

Cluster moves vanish outside their activation region. Entries 7–10 are exactly zero unless *at least two* forward-bond conditions are simultaneously satisfied. In the branch regions where all $A_{ab} = 0$, the 3-particle cluster move has zero probability — the system is purely a single-particle mover in that parameter regime.

5.2 Feasibility of Exhaustive Branch Enumeration (EBE)

EBE replaces the 30-point random substitution of the current SZ check with an explicit test of every feasible branch region. For this orbit rep:

Quantity	Value	Notes
Distinct conditions k (rep 1)	6	$\{A_{12}, A_{13}, A_{23}, B_{12}, B_{13}, B_{23}\}$
Branch regions (2^k)	64	upper bound; some may be infeasible
Leaves contributing to any $T(\mathbf{s} \rightarrow \mathbf{t})$	102	spread across depth 6–10
Distinct target states	27	

Across all 56 orbit representatives, more conditions will appear (different neighbours $\Rightarrow$ different type-pair/distance combinations), but conditions repeat heavily across reps. A conservative estimate gives $k \approx 10$ –15 globally, i.e. 2^{10} – $2^{15} = 1,024$ – $32,768$ regions total.

For each region, the procedure is:

1. Find a rational coupling point $\mathbf{J}^*$ satisfying that region’s linear inequality system (a small LP).
2. Substitute $\mathbf{J}^*$: every Piecewise collapses to either a rational number or a rational $\times e^{-\beta \cdot r}$ with $r \in \mathbb{Q}$.
3. With $\beta = 1$ absorbed, all weights are rationals or rational $\times e^{-r}$ with $r \in \mathbb{Q}$.
4. For each communicating pair (s, t) , form $T(s \rightarrow t)e^{-E(s)} - T(t \rightarrow s)e^{-E(t)}$ and call `$dbcIsExpZero`: exact rational arithmetic, no floating point.

The current random-substitution SZ check with 30 points tests 30 out of 64 (or 2^{15}) regions. With $k = 6$ for rep 1, the probability of any given region being missed is $(1 - 1/64)^{30} \approx 62\%$; for $k = 15$, this rises to essentially 100%. EBE eliminates this coverage gap entirely and replaces a probabilistic certificate with a proof.

The structure of the expressions — products of at most 5 Piecewise factors, with conditions that are simple linear inequalities in at most 4 coupling constants — makes the LP feasibility checks trivially fast ($\ll 1$ ms each), and the `$dbcIsExpZero` check on each pair is a handful of rational additions. **EBE is therefore not only exact but likely faster than the current 30-point SZ loop for this system.**